

UPPER-BODY CONTEXT NODE — HARDWARE ARCHITECTURE, POWER/COMMS, STANDALONE USE CASES, AND OPTIONAL INTEGRATION (NON-LIMITING)

A. Hardware Architecture (Non-Limiting)

In various embodiments, Node 5 comprises an upper-body wearable with a sealed housing containing electronics, a power source, and one or more sensing regions. Node 5 may be implemented as a chest module, ring, wristband, or equivalent form factor. Non-limiting architectural elements include: 1. ECG electrode interfaces (when implemented as chest or other electrode-bearing form factors); 2. an optical sensing module (PPG) with optical coupling features; 3. a 6-axis IMU; 4. EDA/skin conductance electrodes (when implemented as wristband or equivalent); 5. one or more temperature sensors; 6. contact integrity sensing circuitry; 7. communications and security circuitry; and 8. on-device storage for buffering and encrypted recordkeeping.

B. Power, Charging, and Duty Cycles (Non-Limiting)

In various embodiments, Node 5 includes a rechargeable battery and power management circuitry configured for continuous or near-continuous wear. Non-limiting approaches include: 1. low-power sampling modes during baseline monitoring; 2. higher-resolution sampling during selected windows (e.g., sleep or user-requested sessions); 3. docking-based charging and/or inductive charging; and 4. battery health monitoring and safe shutdown behaviors.

C. Secure Communications and On-Device Encryption (Non-Limiting)

In various embodiments, Node 5 implements security controls including: 1. on-device encryption for stored records (data at rest); 2. secure pairing and authenticated encrypted communications (data in transit); 3. firmware integrity and signed update pathways; and 4. access control boundaries separating raw streams from derived summaries. In certain embodiments, Node 5 stores only privacy-preserving context summaries unless raw data retention is explicitly enabled.

D. Standalone Use Cases (Non-Limiting)

In standalone embodiments, Node 5 operates as an independent product to provide privacy-preserving context outputs including: 1. rest/activity and recovery summaries; 2. HR and HRV indices with confidence indicators; 3. respiration rate/variability indices with confidence indicators; 4. autonomic activation indices derived from EDA and HRV; 5. thermal stability and wear integrity indicators; and 6. trend summaries over days and weeks.

E. Optional Multi-Node Integration (Non-Limiting)

In optional multi-node embodiments, Node 5 provides context tags and confidence-weighting that condition other node interpretation. Non-limiting integration behaviors include: 1. selecting matched windows for correlation across nodes under consent; 2. suppressing or down-weighting other node outputs during high-motion or low-wear-integrity windows; 3. providing baseline normalization factors for rest vs. activity contexts; and 4. providing confidence-weighted context conditioning for cross-device correlation outputs.

F. Variations

Any embodiment described herein may omit sensors, add sensors, substitute equivalent modalities, alter form factors, and/or alter processing distribution (on-node, on-device, cloud) while maintaining the functional intent of capturing upper-body context signals and providing context-gated interpretation and confidence-weighted outputs.